

A Multimodal Extension of the YFB Model

Matched expression and methylation: likelihood, empirical-Bayes priors, and variational updates

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1 Purpose and review status

This document adapts the existing YFB model in multiomicsGEP to matched expression and DNA methylation measurements. It is a methodological proposal for advisor review, not a report of a fitted multimodal model. No production fitting code, model defaults, simulations, or real-data analyses are changed.

There are two distinct adaptations to assess. First, partition the features by modality and estimate separate loading priors while sharing the subject scores. Second, derive the survival-feedback version of YFB from a coherent objective. The current default YFB algorithm suppresses survival feedback into the loadings. Its direct multimodal counterpart is retained here as a compatibility baseline. Differences required by a joint derivation are identified separately from differences required simply by adding a modality.

The intended application has **one training cohort** and **one independent validation cohort**, each with matched expression and methylation data. The validation subjects are not additional training subjects. One right-censored survival endpoint is assumed. Outcome-dependent decisions must use training data only.

The starting prior is nonnegative point-exponential for both L and each F_m , with separate fitted parameters for every column and modality. Signed alternatives are derived for comparison. All K score columns are shared, but an entire loading column may vanish in one modality. Ordinary observation likelihood is examined first; explicit modality weighting is an alternative, not an assumed solution.

Choices still requiring advisor review: loading support and analysis scale; ordinary versus weighted objective; raw versus normalized survival projection; and the approximation used for joint inference. The derivations below make those choices reviewable without claiming they have been selected.

2 Current implementation: the reference to be adapted

The source audit refers to repository revision c13eaea, inspected on September 10, 2026. The executable modules take precedence over older companion documents when identifying implemented behavior.

Quantity	R representation	Implemented YFB behavior
Subject scores	EL, EL2	$n \times K$ means and second moments; omics-only update
Feature loadings	EF, EF2	$p \times K$ means and second moments
Survival coefficients	EBeta, EBeta2	K means and second moments for a single cohort
Residual precision	Tau	p point estimates, not a variational distribution
Projection scale	EF_norms	Training loading-column norms used for prediction
Priors	prior_LF, prior_beta	Defaults: point-exponential and Normal, respectively

The relevant entry point is `code/fit_cox_on_yf.R`. Its update modules are `code/update_L_surv_YFB.R`, `code/update_F_surv_YFB.R`, `code/update_beta.R`, and `code/update_tau.R`. Prediction is in `code/predict_cox_on_yf.R`. For reference, the LB model in `code/fit_modular.R` uses $\eta = L\beta$, whereas YFB predicts survival from projections of the observed omics.

The current YFB fit uses

$$\widehat{Z} = Y\bar{F}D^{-1}, \quad D_{kk} = \sqrt{\sum_j \bar{f}_{jk}^2 + 10^{-10}}, \quad \widehat{\eta} = \widehat{Z}\bar{\beta}. \quad (1)$$

Here Z is not L . Even with perfect reconstruction, $YF = LF^\top F$ need not equal L .

The defaults are $\alpha_F = 0$, $\alpha = 0.5$, and $\text{boost_beta} = \text{FALSE}$. Thus L and F use unweighted omics information, while the coefficient update uses α times the survival quadratic. Fitting one prior per column is already the pattern; the multimodal change is to fit one loading prior per **modality and column**, instead of pooling all features into that column’s prior.

2.1 Differences that a joint derivation must not conceal

1. **Feedback:** a Cox factor depending on F contributes to its update. Setting α_F to zero removes that contribution. The resulting procedure is modular; it is not coordinate ascent on the full joint posterior specified below.
2. **Projection uncertainty:** passing $\hat{Z}^2$ as a second moment treats the current estimated projection as fixed. Joint inference on F requires additional variance terms.
3. **Feature interactions:** the existing optional survival contribution in `update_F_surv_YFB.R` uses only the diagonal of $Y^\top W Y$ and a factor-excluded working response. An exact scalar loading update also subtracts the contributions of the other features in that same factor.
4. **Curvature:** the helper `calc_cox_taylor_yf` computes a working curvature described precisely in Section 5. It is not the exact negative diagonal Hessian despite that wording in older comments.
5. **Objective weights:** under the default `per_p` reporting convention, the reported survival contribution is multiplied by p , while the coefficient update is not. Omics updates also do not use the reported factor $(1 - \alpha)$. Therefore the reported combined diagnostic is not a single objective whose coordinate updates reproduce all defaults.
6. **Normalization:** the optional survival loading update is written for a linear, unnormalized projection; differentiating a normalized projection requires additional terms.

These observations motivate explicit comparisons. They are not changes to the established single-modality implementation.

3 Data, parameterization, and priors

3.1 Matched blocks and notation

For modality $m \in \{1, \dots, M\}$, let $Y_m \in \mathbb{R}^{n \times p_m}$. Initially $M = 2$: expression and methylation. Row i refers to the same subject in every block. Each subject has observed follow-up $t_i = \min(T_i, C_i)$ and event indicator $\delta_i = 1\{T_i \leq C_i\}$. The likelihood below conditions on the realized preprocessing and feature selection; fitting those operations is a separate training-only step.

$$\begin{aligned} p &= \sum_{m=1}^M p_m, & Y &= [Y_1 \ \dots \ Y_M] \in \mathbb{R}^{n \times p}, \\ F &= [F_1^\top \ \dots \ F_M^\top]^\top \in \mathbb{R}^{p \times K}, & L &\in \mathbb{R}^{n \times K}. \end{aligned} \tag{2}$$

Use k for a factor, j for a feature within modality m , and $a = (m, j)$ for a flattened feature index. Gene and CpG identifiers occupy distinct namespaces. No gene-to-CpG pairing is required.

The Gaussian observation model is

$$y_{mij} \mid L, F_m, \tau_{mj} \stackrel{\text{ind}}{\sim} \mathcal{N} \left(\sum_{k=1}^K l_{ik} f_{mjk}, \tau_{mj}^{-1} \right). \tag{3}$$

Conditional residual independence across features and modalities is an assumption, not an assertion that CpGs are marginally independent. Shared factors induce marginal dependence. Residual dependence left after factorization is a potential source of overconfidence at large p_m . Residual precisions vary across features within each modality. A single modality-wide precision is not introduced.

For derivation of the linear YFB case, define **fixed** projection coefficients c_m and the columns

$$x_{mj} = c_m Y_m[:, j], \quad X = [c_1 Y_1 \ \cdots \ c_M Y_M], \quad z_k = X f_k, \quad \eta = \sum_k \beta_k z_k. \quad (4)$$

The unweighted stacked projection has $c_m = 1$ and $Z = YF = \sum_m Y_m F_m$. Coefficients c_m change prediction, whereas the likelihood weights introduced later change estimation of the reconstruction. They are separate choices. In the core derivation they are fixed constants, not parameters silently re-estimated at each iteration.

3.2 Score and loading priors

Use conditionally independent entries with column-specific distributions:

$$l_{ik} \sim g_{L_k}, \quad f_{mjk} \sim g_{F_{mk}}, \quad \beta_k \sim g_{\beta_k}. \quad (5)$$

The starting score and loading families are

$$\begin{aligned} g_{L_k} &= (1 - \pi_{L_k}) \delta_0 + \pi_{L_k} \text{Exp}(r_{L_k}), \\ g_{F_{mk}} &= (1 - \pi_{F_{mk}}) \delta_0 + \pi_{F_{mk}} \text{Exp}(r_{F_{mk}}). \end{aligned} \quad (6)$$

Here π is the **probability of the nonzero component**, $r > 0$ is an exponential rate, and δ_0 is a unit point mass at zero. The package scale parameter is $1/r$, not r . The prior mode remains fixed at zero. There are $2K$ score-prior hyperparameters and $2MK$ loading-prior hyperparameters before considering boundary cases. No additional hyperpriors are imposed: these are empirical-Bayes estimates.

The minimal signed comparison replaces $g_{F_{mk}}$ by

$$g_{F_{mk}} = (1 - \pi_{F_{mk}}) \delta_0 + \pi_{F_{mk}} \mathcal{N}(0, s_{F_{mk}}^2). \quad (7)$$

It can be applied to both modalities or to a prespecified subset. The score family can remain point-exponential for that comparison. A fully signed alternative also replaces g_{L_k} by point-normal; this is a separate modeling choice, particularly relevant for centered data. The updates below support either family.

The coefficient baseline retains the implemented zero-centered Normal family:

$$\beta_k \sim \mathcal{N}(0, s_{\beta_k}^2). \quad (8)$$

Each variance is fitted empirically. With only one coefficient per factor, this is a weakly informed, boundary-prone empirical-Bayes problem; it is not pooling across K coefficients. A common variance or spike-and-slab coefficient family would be an additional choice, not a hidden default. The document gives the Normal and spike-and-slab algebra so that such a comparison is possible.

3.3 What shared means

Sharing means the same l_{ik} participates in every modality's reconstruction. It does not require $\pi_{Fmk} > 0$ or a nonzero posterior column in every modality. A zero block in methylation leaves an expression-only factor within the common K -column space. Separate priors do not introduce different subject scores for each modality, and no cross-modality prior ties f_{1jk} to a particular CpG loading.

4 From joint likelihood to the variational objective

4.1 Observation and survival likelihoods

The complete Gaussian observation log-likelihood is

$$\ell_m = -\frac{np_m}{2} \log(2\pi) + \frac{n}{2} \sum_j \log \tau_{mj} - \frac{1}{2} \sum_{i,j} \tau_{mj} \left(y_{mij} - \sum_k l_{ik} f_{mjk} \right)^2. \quad (9)$$

Under proportional hazards, $h_i(t) = h_0(t) \exp(\eta_i)$. Conditional on independent, noninformative censoring, the event-time likelihood contribution, omitting censoring factors, is

$$\mathcal{L}_T(h_0, \eta) = \prod_i \{h_0(t_i) e^{\eta_i}\}^{\delta_i} \exp\{-H_0(t_i) e^{\eta_i}\}. \quad (10)$$

Consequently, a full joint likelihood with a specified baseline is $\{\prod_m p(Y_m | L, F_m, \tau_m)\} \mathcal{L}_T(h_0, \eta)$. Multiplying by the priors gives the corresponding posterior kernel. This factorization is coherent even though Y also enters η : it specifies $p(Y | L, F, \tau) p(T | Y, F, \beta, h_0)$.

The existing method eliminates the baseline hazard from coefficient estimation using the Cox partial likelihood. At distinct event time u_r , write $\mathcal{D}_r = \{i : t_i = u_r, \delta_i = 1\}$, $d_r = |\mathcal{D}_r|$, and $\mathcal{R}_r = \{i : t_i \geq u_r\}$. With the Breslow convention for ties,

$$\ell_{\text{Cox}}(\eta) = \sum_r \left\{ \sum_{i \in \mathcal{D}_r} \eta_i - d_r \log \sum_{i \in \mathcal{R}_r} e^{\eta_i} \right\}. \quad (11)$$

This is the adopted survival estimation factor. It is not a normalized joint density of event times. For untied events its conditional risk-set interpretation motivates partial-likelihood inference; Breslow specifies the tie approximation. Absolute survival probabilities would additionally need an estimated baseline, for example $\Delta \widehat{H}_0(u_r) = d_r / \sum_{i \in \mathcal{R}_r} e^{\eta_i}$. The present target is relative risk, not absolute survival calibration.

4.2 Ordinary and weighted estimation targets

Let $a_m > 0$ be an omics-likelihood weight and $b \geq 0$ a survival weight. The generalized posterior kernel is

$$\tilde{p}(L, F, \beta | Y, t, \delta; \tau, g) \propto \exp \left\{ \sum_m a_m \ell_m + b \ell_{\text{Cox}}(\eta) \right\} \prod_{i,k} g_{L_k}(l_{ik}) \prod_{m,j,k} g_{F_{mk}}(f_{mjk}) \prod_k g_{\beta_k}(\beta_k). \quad (12)$$

The ordinary observation-likelihood reference is $a_m = 1$, $b = 1$. Even there the survival factor is a partial likelihood. Nonunit weights define a power-likelihood or generalized-Bayes target, not the original sampling likelihood. Weights are prespecified or tuned inside training; there is no weight update in this document. Setting $b = 0$ removes survival from the *entire joint target*, including β ; it does not reproduce the no-feedback baseline, which still fits β .

Use the mean-field family

$$q(L, F, \beta) = \prod_{i,k} q(l_{ik}) \prod_{m,j,k} q(f_{mjk}) \prod_k q(\beta_k). \quad (13)$$

Define means $\bar{l}, \bar{f}, \bar{\beta}$, second moments $l^{(2)}, f^{(2)}, \beta^{(2)}$, and variances $v_l = l^{(2)} - \bar{l}^2$, $v_f = f^{(2)} - \bar{f}^2$, $v_\beta = \beta^{(2)} - \bar{\beta}^2$. The second moments are not squared means. The exact variational functional for the kernel is

$$\begin{aligned} \mathcal{J}(q, \tau, g) = & \sum_m a_m \mathbb{E}_q \ell_m + b \mathbb{E}_q \ell_{\text{Cox}}(\eta) \\ & - \sum_{i,k} \text{KL}(q(l_{ik}) \| g_{L_k}) \\ & - \sum_{m,j,k} \text{KL}(q(f_{mjk}) \| g_{F_{mk}}) - \sum_k \text{KL}(q(\beta_k) \| g_{\beta_k}). \end{aligned} \quad (14)$$

For fixed hyperparameters it bounds the log normalizing constant of the kernel. It is not automatically a bound on a full event-time marginal likelihood. Empirical Bayes optimizes this functional over g and τ as well as q . Mixed discrete/continuous priors require KL with respect to the appropriate mixed measure; a Gaussian differential entropy alone would omit the point mass.

The expectation of the Cox log-sum-exp has no general closed form. The following updates optimize a stated quadratic surrogate. Keeping that approximation visible is essential.

5 Cox expansion and projection uncertainty

5.1 Exact derivatives and the implemented working curvature

At an expansion point η_0 , define the n -vector of risk-set probabilities P_r by

$$(P_r)_i = \frac{1\{i \in \mathcal{R}_r\} e^{\eta_{0i}}}{\sum_{h \in \mathcal{R}_r} e^{\eta_{0h}}}.$$

Differentiating Equation 11 gives

$$u = \delta - \sum_r d_r P_r, \quad H = -\nabla^2 \ell_{\text{Cox}}(\eta_0) = \sum_r d_r \{\text{diag}(P_r) - P_r P_r^\top\}. \quad (15)$$

H is positive semidefinite and generally dense. Its rows sum to zero, reflecting invariance to a common shift in η . Its diagonal is $\sum_r d_r (P_r)_i \{1 - (P_r)_i\}$.

The current helper instead uses, before its numerical floor,

$$W_{\text{work}} = \text{diag} \left(\sum_r d_r P_r \right). \quad (16)$$

This omits the negative outer-product terms, including their diagonals. At η_0 , $W_{\text{work}} - H = \sum_r d_r P_r P_r^\top$ is positive semidefinite. This local relation does **not** establish a global quadratic lower bound at all η . The implemented floor is 10^{-6} on each working weight; it is a numerical alteration, not part of the exact Hessian.

5.2 A common notation for the quadratic surrogate

For either $W = H$ or a stated working approximation, write

$$Q(\eta; \eta_0, W) = \ell_{\text{Cox}}(\eta_0) + u^\top(\eta - \eta_0) - \frac{1}{2}(\eta - \eta_0)^\top W(\eta - \eta_0). \quad (17)$$

Define

$$t^* = u + W\eta_0, \quad C_0 = \ell_{\text{Cox}}(\eta_0) - u^\top \eta_0 - \frac{1}{2}\eta_0^\top W\eta_0.$$

Then $Q = C_0 + (t^*)^\top \eta - \eta^\top W\eta/2$. For diagonal positive W , the familiar working response is $z^* = \eta_0 + W^{-1}u$, so $t^* = Wz^*$. The t^* form also works with the singular full Hessian and avoids an unnecessary matrix inverse.

Replace $\mathbb{E}\ell_{\text{Cox}}$ in Equation 14 by $\mathbb{E}Q$ to obtain $\mathcal{J}_Q$. The expansion point, u , and W are held fixed during an inner coordinate-ascent cycle. In particular, the linear coefficient is $u + W\eta_0$, not just u .

5.3 Moments of projected scores and the risk predictor

Let $\bar{z}_k = X\bar{f}_k$ and let $v_{f,ak}$ be the loading variance. Mean-field independence of distinct loadings gives

$$V_{z_k} = \text{Cov}(z_k) = \sum_a v_{f,ak} x_a x_a^\top, \quad \mathbb{E}[z_k z_k^\top] = \bar{z}_k \bar{z}_k^\top + V_{z_k}. \quad (18)$$

Although individual loading posteriors are independent, different subjects' projections are correlated: the same uncertain loadings are used for every subject. Distinct factor columns are independent under the chosen variational family. Thus

$$\begin{aligned} \mu_\eta &= \sum_k \bar{\beta}_k \bar{z}_k, \\ V_\eta &= \sum_k \left\{ \beta_k^{(2)} V_{z_k} + v_{\beta,k} \bar{z}_k \bar{z}_k^\top \right\}, \\ \mathbb{E}Q &= C_0 + (t^*)^\top \mu_\eta - \frac{1}{2} \mu_\eta^\top W \mu_\eta - \frac{1}{2} \text{tr}(W V_\eta). \end{aligned} \quad (19)$$

At $\eta_0 = \mu_\eta$, the last expression simplifies to $\ell_{\text{Cox}}(\mu_\eta) - \text{tr}(W V_\eta)/2$. After inner updates move μ_η , the unsimplified expression must be used until the expansion is refreshed. Mixing an old log partial likelihood with updated means and dropping the linear correction evaluates a different quantity.

For diagonal W , only $\text{Var}(\eta_i)$ is needed, but the $\beta_k^{(2)} V_{z_k}$ term remains. Dropping it is precisely the fixed-projection approximation.

6 Coordinate updates from the surrogate objective

6.1 The normal-means reduction

For any scalar coordinate θ with prior g , suppose its expected log-likelihood contribution, holding other variational factors fixed, is

$$-\frac{1}{2}A\theta^2 + B\theta + \text{constant}, \quad A > 0.$$

The variational functional for this coordinate is the expectation of this expression plus $\mathbb{E}_q \log g - \mathbb{E}_q \log q$. Taking its functional derivative subject to $\int q = 1$ yields

$$\log q^*(\theta) = \log g(\theta) - \frac{1}{2}A\theta^2 + B\theta + C.$$

Completing the square with $x = B/A$ and $s = A^{-1/2}$ gives

$$q^*(d\theta) = \frac{\mathcal{N}(x; \theta, s^2)g(d\theta)}{\int \mathcal{N}(x; v, s^2)g(dv)}. \quad (20)$$

This establishes the EBNM interface; it is not a substitute for deriving A and B for each parameter. The symbol B here is a linear coefficient in a quadratic, not the vector of survival coefficients. Throughout this document the user's survival B is denoted β .

6.2 Shared subject scores L

For factor k , define the reconstruction residual excluding that factor:

$$\bar{R}_{mij}^{-k} = y_{mij} - \sum_{h \neq k} \bar{l}_{ih} \bar{f}_{mjh}. \quad (21)$$

Holding the other coordinates fixed, expand the observation term:

$$\begin{aligned} & -\frac{1}{2} \sum_{m,j} a_m \tau_{mj} \mathbb{E}_{-l_{ik}} [(R_{mij}^{-k} - l_{ik} f_{mjk})^2] \\ & = -\frac{1}{2} l_{ik}^2 \sum_{m,j} a_m \tau_{mj} f_{mjk}^{(2)} + l_{ik} \sum_{m,j} a_m \tau_{mj} \bar{R}_{mij}^{-k} \bar{f}_{mjk} + C. \end{aligned}$$

The residual is independent of f_{mjk} under mean field because it excludes factor k . Under YFB, η contains F and Y , not L , so there is no survival contribution to this coordinate. Therefore

$$\begin{aligned} A_{L,ik} &= \sum_m a_m \sum_j \tau_{mj} f_{mjk}^{(2)}, \\ B_{L,ik} &= \sum_m a_m \sum_j \tau_{mj} \bar{R}_{mij}^{-k} \bar{f}_{mjk}. \end{aligned} \quad (22)$$

The precision is the same across subjects with complete modalities. Apply Equation 20 with g_{L_k} . An n -observation EBNM fit per factor estimates its shared score prior. Return $\bar{l}_{ik} = \mathbb{E} l_{ik}$ and $l_{ik}^{(2)} = \text{Var}(l_{ik}) + \bar{l}_{ik}^2$. Compared with current YFB, the only structural change is the sum over modalities, with explicit weights when used.

6.3 Survival coefficients β

Define the mean contribution of all other factors to risk:

$$\mu_{\eta,-k} = \sum_{h \neq k} \bar{\beta}_h \bar{z}_h, \quad h_k = t^* - W \mu_{\eta,-k}. \quad (23)$$

Insert $\eta = \eta_{-k} + \beta_k z_k$ into the quadratic. Terms depending on β_k are

$$b\beta_k \bar{z}_k^\top h_k - \frac{b}{2}\beta_k^2 \mathbb{E}[z_k^\top W z_k].$$

The cross term factors because η_{-k} and z_k involve different mean-field factor columns. By Equation 18,

$$\mathbb{E}[z_k^\top W z_k] = \bar{z}_k^\top W \bar{z}_k + \text{tr}(W V_{z_k}).$$

Thus

$$\begin{aligned} A_{\beta,k} &= b \left\{ \bar{z}_k^\top W \bar{z}_k + \sum_a v_{f,ak} x_a^\top W x_a \right\}, \\ B_{\beta,k} &= b \bar{z}_k^\top h_k. \end{aligned} \tag{24}$$

With a Normal prior variance $s_{\beta k}^2$, completing the square including its precision gives

$$v_{\beta,k} = (A_{\beta,k} + s_{\beta k}^{-2})^{-1}, \quad \bar{\beta}_k = v_{\beta,k} B_{\beta,k}, \quad \beta_k^{(2)} = v_{\beta,k} + \bar{\beta}_k^2. \tag{25}$$

For a spike-and-slab alternative, use the posterior formula in Section 7. The extra trace term in Equation 24 is the loading-uncertainty correction missing from a fixed-projection fit. For diagonal W and $v_f = 0$, these equations reduce to the implemented beta-update algebra with $b = \text{alpha}/\text{beta_divisor}$.

6.4 Modality-specific loadings F_m

6.4.1 Reconstruction contribution

For $a = (m, j)$ and factor k , expand

$$\begin{aligned} & -\frac{a_m \tau_{mj}}{2} \sum_i \mathbb{E}_{-f_{mjk}} (R_{mij}^{-k} - l_{ik} f_{mjk})^2 \\ & = -\frac{1}{2} A_{F,\text{om},ak} f_{mjk}^2 + B_{F,\text{om},ak} f_{mjk} + C, \end{aligned}$$

where

$$A_{F,\text{om},ak} = a_m \tau_{mj} \sum_i l_{ik}^{(2)}, \quad B_{F,\text{om},ak} = a_m \tau_{mj} \sum_i \bar{l}_{ik} \bar{R}_{mij}^{-k}. \tag{26}$$

Only this feature's modality appears in its reconstruction likelihood. Its shared score column carries the information integrated from the other modalities.

6.4.2 Survival contribution, including same-factor features

Write

$$z_k = x_a f_{ak} + z_{k,-a}, \quad \bar{z}_{k,-a} = \bar{z}_k - x_a \bar{f}_{ak}.$$

The factor- k part of the expected survival quadratic is

$$b \bar{\beta}_k \mathbb{E}[z_k]^\top h_k - \frac{b}{2} \beta_k^{(2)} \mathbb{E}[z_k^\top W z_k] + C.$$

Expanding the second term while keeping the scalar f_{ak} free gives

$$\mathbb{E}_{-f_{ak}} [z_k^\top W z_k] = f_{ak}^2 x_a^\top W x_a + 2 f_{ak} x_a^\top W \bar{z}_{k,-a} + C.$$

Consequently,

$$\begin{aligned} A_{F,\text{surv},ak} &= b\beta_k^{(2)}x_a^\top Wx_a, \\ B_{F,\text{surv},ak} &= b\left\{\bar{\beta}_kx_a^\top h_k - \beta_k^{(2)}x_a^\top W\bar{z}_{k,-a}\right\}. \end{aligned} \quad (27)$$

The second term in $B_{F,\text{surv},ak}$ subtracts the contribution of **all other features in the same factor**, including the other modality. It has $\beta_k^{(2)}$, not $\bar{\beta}_k^2$, because the same random coefficient multiplies both contributions. Removing factor k from the working response is not enough to remove these interactions. Even a diagonal W does not make $X^\top WX$ diagonal.

Combine the two sources:

$$A_{F,ak} = A_{F,\text{om},ak} + A_{F,\text{surv},ak}, \quad B_{F,ak} = B_{F,\text{om},ak} + B_{F,\text{surv},ak}. \quad (28)$$

Apply Equation 20 with the prior $g_{F_{mk}}$ belonging to this modality and factor. The priors remain separate even though the survival contributions couple their posterior updates.

6.4.3 Consequence for implementation

In omics-only factorization, the feature updates within a factor are conditionally separable, so one vectorized EBNM call can update a whole loading block and estimate its prior. In joint YFB, $B_{F,ak}$ depends on the current means of the other features. A simultaneous update using stale values is a parallel fixed-point step, not an exact block coordinate maximization.

A direct derivation-faithful algorithm therefore performs scalar Gauss-Seidel posterior updates with the current modality prior fixed, maintains $\bar{z}_k$ incrementally, and updates the prior parameters using all features in that modality after a sweep. This is a variational-EM procedure, not one independently re-fitted prior per feature. A faster block or proximal method may later be designed, but its objective and convergence behavior must be checked separately.

6.5 Residual precisions τ_{mj}

Independence of factor columns gives the variance-corrected residual moment

$$S_{mij} = \left(y_{mij} - \sum_k \bar{l}_{ik} \bar{f}_{mjk}\right)^2 + \sum_k \left\{l_{ik}^{(2)} f_{mjk}^{(2)} - \bar{l}_{ik}^2 \bar{f}_{mjk}^2\right\}. \quad (29)$$

To obtain it, expand the squared sum of factor products. Cross-factor expectations equal products of means; the difference between the second moment and the squared mean survives only within each factor. This explains both the plus sign and the necessity of second moments.

The terms involving τ_{mj} are

$$\frac{a_m n}{2} \log \tau_{mj} - \frac{a_m \tau_{mj}}{2} \sum_i S_{mij}.$$

Differentiate and set the result to zero:

$$\frac{a_m n}{2\tau_{mj}} - \frac{a_m}{2} \sum_i S_{mij} = 0 \quad \Rightarrow \quad \hat{\tau}_{mj} = \frac{n}{\sum_i S_{mij}}. \quad (30)$$

The second derivative is negative for $a_m > 0$. A constant modality weight cancels from this point-estimation update; it does not cancel from the score and loading posterior precisions. If a modality

is assigned weight zero, its residual precisions are not identified by this objective. Positive weights are assumed throughout.

The existing denominator floor gives $\tau_{mj} \leq 1/\text{tau_floor}$ and should be described as a constrained estimate. The multimodal method can reuse the algebra blockwise without allocating all S_m simultaneously. No Gamma-prior, shape/rate update, or $q(\tau)$ is implied.

7 Prior estimation and posterior moments

7.1 Generic empirical-Bayes step

For a conditionally separable block with pseudo-data (x_d, s_d) , optimizing its posterior factors and common prior jointly gives

$$\hat{g} = \arg \max_{g \in \mathcal{G}} \sum_d \log \left\{ \int \mathcal{N}(x_d; \theta, s_d^2) g(d\theta) \right\}. \quad (31)$$

This follows by substituting the normalized posterior Equation 20 into the block variational objective: the posterior-versus-prior and pseudo-likelihood terms reduce to the log normalizing integral. Apply this to n subjects for g_{L_k} , to p_m features for $g_{F_{mk}}$ in the no-feedback case, and to the scalar coefficient for g_{β_k} .

In coupled joint loading updates, the pseudo-data change when other features change. Maximizing a product of marginal densities computed from stale pseudo-data is not an exact joint loading-block update. Instead alternate fixed-prior coordinate posteriors with the fixed-posterior hyperparameter step

$$\hat{g} = \arg \max_g \sum_d \mathbb{E}_{q_d} \log g(\theta_d). \quad (32)$$

The closed-form steps below implement this variational-EM option. They also make every proposed hyperparameter estimable.

7.2 Point-normal family

For prior $(1 - \pi)\delta_0 + \pi\mathcal{N}(0, s_0^2)$ and pseudo-likelihood $x \mid \theta \sim \mathcal{N}(\theta, s^2)$, define

$$d_0 = \mathcal{N}(x; 0, s^2), \quad d_1 = \mathcal{N}(x; 0, s^2 + s_0^2), \quad r = \frac{\pi d_1}{(1 - \pi)d_0 + \pi d_1}.$$

Conditional on being in the slab, completing the Gaussian square gives

$$v = (s^{-2} + s_0^{-2})^{-1}, \quad \mu = vx/s^2.$$

The posterior moments are

$$\mathbb{E}\theta = r\mu, \quad \mathbb{E}\theta^2 = r(v + \mu^2). \quad (33)$$

Introduce a slab indicator γ_d . The expected complete-prior log density for N coordinates is, up to constants,

$$\sum_d \{r_d \log \pi + (1 - r_d) \log(1 - \pi)\} - \frac{1}{2} \sum_d r_d \log s_0^2 - \frac{1}{2s_0^2} \sum_d \mathbb{E}\theta_d^2.$$

Differentiating with respect to π and s_0^2 yields

$$\pi_{\text{new}} = \frac{1}{N} \sum_d r_d, \quad s_{0,\text{new}}^2 = \frac{\sum_d \mathbb{E}\theta_d^2}{\sum_d r_d}. \quad (34)$$

The numerator is the unconditional second moment because the spike contributes zero. These equations apply separately to each selected modality-factor prior; no sum over the other modality appears.

7.3 Point-exponential family

Use $(1 - \pi)\delta_0 + \pi r_0 e^{-r_0\theta} 1\{\theta \geq 0\}$. Multiplying the slab density by the normal pseudo-likelihood and completing the square yields a Normal distribution truncated below at zero with underlying mean $\mu = x - r_0 s^2$ and variance s^2 . Writing $h = \mu/s$ and $\lambda(h) = \phi(h)/\Phi(h)$,

$$\begin{aligned} d_1 &= r_0 \exp(-r_0 x + r_0^2 s^2 / 2) \Phi(h), \\ r &= \frac{\pi d_1}{(1 - \pi) \mathcal{N}(x; 0, s^2) + \pi d_1}, \\ M_1 &= \mu + s \lambda(h), \\ M_2 &= \mu^2 + s^2 + \mu s \lambda(h), \\ \mathbb{E}\theta &= r M_1, \quad \mathbb{E}\theta^2 = r M_2. \end{aligned} \quad (35)$$

The slab integral d_1 follows by integrating the completed Normal kernel from zero to infinity. The moment formulas follow from the first two moments of that truncated Normal. Log-scale evaluation of d_1 , Φ , and the inverse Mills ratio is needed in extreme tails; the displayed expressions specify the mathematics, not a recommendation to evaluate unstable exponentials directly.

The complete-prior expected log density contains

$$\sum_d \{r_d \log \pi + (1 - r_d) \log(1 - \pi)\} + \left(\sum_d r_d \right) \log r_0 - r_0 \sum_d \mathbb{E}\theta_d.$$

Setting its derivatives to zero gives

$$\pi_{\text{new}} = N^{-1} \sum_d r_d, \quad r_{0,\text{new}} = \frac{\sum_d r_d}{\sum_d \mathbb{E}\theta_d}. \quad (36)$$

The score prior uses $N = n$; the loading prior uses $N = p_m$. This is where modality-specific sparsity and amplitude are learned separately.

7.4 Normal coefficient family and boundary cases

For a zero-centered Normal prior shared by a block of N coefficients, Equation 32 gives $s_{\beta,\text{new}}^2 = N^{-1} \sum_d \mathbb{E}\beta_d^2$. The implemented factor-specific baseline has $N = 1$. If optimizing the scalar normal-means marginal likelihood jointly instead, differentiation of $-\log(s^2 + s_\beta^2)/2 - x^2/\{2(s^2 + s_\beta^2)\}$ gives

$$\hat{s}_\beta^2 = \max(0, x^2 - s^2).$$

The direct marginal maximizer and one EM step are different operations; repeated EM steps target the same stationary problem in this separable case.

If $\sum r_d = 0$, the fitted prior is the point mass at zero and the slab scale is unidentifiable. Do not divide zero by zero or report a meaningful slab estimate. If $A = 0$ and $B = 0$, the likelihood supplies no information and the fixed-prior posterior equals the prior. A precision floor used for software compatibility is not evidence of information. If all fitted factors vanish or the cohort has no events, the algorithm must report that state rather than claim a prognostic model. Boundary priors can be absorbing under EM, so initial priors should start inside their admissible parameter space and multiple starts should be considered after approval.

7.5 Computing KL contributions correctly

For a posterior obtained from Equation 20 at fixed g , let $m(x) = \int \mathcal{N}(x; \theta, s^2) g(d\theta)$. Then

$$-\text{KL}(q\|g) = \log m(x) - \mathbb{E}_q \log \mathcal{N}(x; \theta, s^2),$$

where

$$\mathbb{E}_q \log \mathcal{N}(x; \theta, s^2) = -\frac{1}{2} \log(2\pi s^2) - \frac{x^2 - 2x\mathbb{E}\theta + \mathbb{E}\theta^2}{2s^2}. \quad (37)$$

This is the identity used by `compute_ebnm_kl`. The sum over loading blocks must retain every modality's term. After a hyperparameter update changes g while holding q fixed, an old EBNM normalizing constant cannot be used unchanged. Evaluate the mixed-distribution KL directly, or add $\mathbb{E}_q \log(g_{\text{new}}/g_{\text{old}})$ to the stored negative KL. Likewise, never recompute pseudo-data and combine their likelihood normalizer with posterior moments from a different update.

8 Dimensional imbalance: what separate priors can and cannot do

8.1 Ordinary likelihood first

With $a_m = 1$, a modality contributes more independent observations when it has more features. Its contribution to score precision is

$$A_{L,k}^{(m)} = \sum_{j=1}^{p_m} \tau_{mj} f_{mjk}^{(2)}.$$

If those expected precision-weighted second moments have comparable means across modalities, $A_{L,k}^{(m)}$ grows with p_m . This is a likelihood property, not a consequence of mistakenly sharing the priors. Separate $g_{F_{mk}}$ can learn different sparsity and amplitude, and feature-wise τ_{mj} can reflect different residual variances. Neither guarantees cancellation of the feature-count effect.

Consider a representative loading prior with nonzero probability π_m . Its second moment is $\pi_m s_m^2$ for point-normal and $2\pi_m/r_m^2$ for point-exponential. At comparable residual precision, a rough prior scale for a modality's contribution is therefore proportional to $p_m \pi_m s_m^2$ or $2p_m \pi_m/r_m^2$. This is an explanatory scale calculation, not the posterior update. Balancing those quantities would require an explicit constraint or hyperprior; independent EB fitting does not impose one.

8.2 Explicit likelihood-weight alternatives

Three choices illustrate the distinction:

Formulation	Omics weights	Interpretation
Ordinary reference	$a_m = 1$	Every measured feature contributes its ordinary likelihood
Per-feature average	$a_m = 1/p_m$	Each modality contributes its mean feature log-likelihood
Equal effective counts	$a_m = p_{\text{ref}}/p_m$	Equal feature-count scale with an explicit reference count

The latter two have the same *relative* modality weights but different absolute evidence strength relative to priors, entropy, and the survival term. The reference p_{ref} is an additional declared constant; it must not be chosen implicitly.

More generally one can write $a_m = \rho_m/p_m$ with fixed positive ρ_m . If the earlier alpha convention is desired, explicitly define $a_m = (1 - \alpha)\rho_m/p_m$ and $b = \alpha/s_T$ with a declared survival scale s_T . This changes every relevant update, not only the reported objective. The ordinary reference is easier to express directly by $a_m = b = 1$; it is not equivalent to $\alpha = 0.5$ when prior/entropy terms are left unchanged.

In the pure-reconstruction loading update, a_m cancels from $x = B/A$ but changes $s = A^{-1/2}$. Consequently it changes shrinkage and the fitted prior. In L , relative modality weights change both x and s . In joint F , cancellation generally fails because survival contributes additional terms. In τ , the weight cancels as shown above.

The loading prior/entropy contribution still contains p_m terms. Weighting the observation likelihood alone is therefore not a theorem of equal posterior modality influence. If one also weights loading KL terms, the coordinate posterior and hyperparameter objective change again; that is a different functional and is not adopted here. Multiplying the **entire** objective by a common positive constant leaves its maximizer unchanged. Multiplying only likelihood terms does not.

8.3 Informative, null, duplicate, and rescaled features

Additional informative features. Under conditional independence they can improve estimation of shared scores. Removing all benefit of extra informative measurements is not necessarily desirable. Evaluate whether the larger block contributes independent signal or redundant evidence.

Additional null features. An adaptive loading prior can shrink their means, but posterior second moments can remain positive. Their aggregate precision contribution and the fitted noise levels should be inspected. A large CpG block must not be assumed harmless merely because most posterior means are small.

Duplicated features. If one modality's features and loading moments are repeated r times, its unweighted score precision and signal multiply by r . With weights divided by r , these particular sums are unchanged. The full fit need not be invariant: there are now more prior terms and correlated residuals, and YFB projection contributions also duplicate unless separately adjusted.

This is a useful diagnostic of pseudoreplication, not an exact invariance requirement for the entire weighted model.

Measurement rescaling. Under $Y'_m = d_m Y_m$, $F'_m = d_m F_m$, and $\tau'_m = \tau_m/d_m^2$, the Gaussian model transforms consistently, with the usual data-density Jacobian. Raw YFB projections instead satisfy $Y'_m F'_m = d_m^2 Y_m F_m$. Thus noise adjustment alone does not make YFB prediction scale-invariant. Modality preprocessing and projection normalization need their own specification.

The initial scientific comparison should keep the ordinary likelihood, fit separate priors, and examine these contributions. Weighted targets are explicit alternatives if the resulting domination reflects unwanted feature-count or redundancy effects. The comparison must use training data; the independent cohort cannot decide the weights.

9 Projection normalization, support, and identifiability

9.1 Three projection definitions

For a point estimate of the loading columns, compare

$$\begin{aligned} \text{raw: } z_k &= \sum_m c_m Y_m f_{mk}, \\ \text{global norm: } z_k &= \frac{\sum_m Y_m f_{mk}}{\sqrt{\sum_m \|f_{mk}\|^2}}, \\ \text{per-modality norm: } z_k &= \sum_m c_m Y_m \frac{f_{mk}}{\|f_{mk}\|}. \end{aligned} \tag{38}$$

The first is the linear model derived in full above. The second matches the structure of current YFB prediction, apart from its numerical epsilon. The third gives each modality its own unit-norm loading direction; it is an alternative for review, not a guaranteed balance of projected variance. Correlation and variance within Y_m still matter. It can also amplify a very weak nonzero loading block, making the zero-block convention particularly consequential.

Current YFB normalizes the **posterior mean** loading, not every draw of F . In general, $\mathbb{E}[F/\|F\|] \neq \mathbb{E}[F]/\|\mathbb{E}[F]\|$. Treating a normalized mean projection as fixed during β fitting is a plug-in approximation. It must not be substituted for a stochastic normalized projection in a joint posterior without stating that change.

9.2 Derivatives for a normalized loading direction

For one column write $r = (f^\top f + \epsilon)^{1/2}$ and $z = Xf/r$. The current epsilon is a numerical regularizer; with $\epsilon > 0$ it changes exact scale invariance. For coordinate f_a ,

$$\begin{aligned} \frac{\partial z}{\partial f_a} &= \frac{x_a}{r} - \frac{(Xf)f_a}{r^3}, \\ \frac{\partial^2 z}{\partial f_a^2} &= -\frac{2x_a f_a}{r^3} - \frac{Xf}{r^3} + \frac{3(Xf)f_a^2}{r^5}. \end{aligned} \tag{39}$$

These follow from $\partial r/\partial f_a = f_a/r$ and the product rule. For per-modality normalization the same equations apply within each block, multiplied by c_m ; for global normalization the norm uses all modalities, so even its denominator couples the blocks.

For a point coefficient β_k , the survival gradient for f_a is $(\partial\eta/\partial f_a)^\top u$. Its second derivative is

$$-\left(\frac{\partial\eta}{\partial f_a}\right)^\top H\left(\frac{\partial\eta}{\partial f_a}\right) + u^\top \frac{\partial^2\eta}{\partial f_a^2}.$$

The second term is absent from a linear projection. It can change concavity. Replacing x_a by x_a/r in the raw update and ignoring the derivatives of r does not derive the normalized model.

9.3 Estimation implications for normalized alternatives

For a stochastic normalized predictor, the exact coordinate equation is

$$q^*(f_a) \propto g(f_a) \exp\left\{-\frac{1}{2}A_{\text{om}}f_a^2 + B_{\text{om}}f_a + b\mathbb{E}_{-f_a}\ell_{\text{Cox}}(\eta(f_a, F_{-a}, \beta))\right\}. \quad (40)$$

The Cox term can instead be replaced by Q for a stated surrogate, but its composition with the normalization remains nonquadratic. L and τ retain their reconstruction updates. For β under a quadratic Cox surrogate, Equation 24 still holds with $\bar{z}_k = \mathbb{E}z_k$ and $V_{z_k} = \text{Cov}(z_k)$ computed for the chosen nonlinear map; Equation 18 is no longer valid. Prior hyperparameter EM steps retain their form once coordinate responsibilities and moments are known.

A numerical implementation of this alternative would integrate the continuous slab and point mass separately for each coordinate, normalize their masses, and integrate their first two moments. Inner expectations over other coordinates require quadrature in tiny checks and controlled Monte Carlo at realistic dimensions. Numerical integration error and Monte Carlo error must be reported. This is substantially more expensive than the linear surrogate algorithm and is not a drop-in EBNM call. A local linearization is a possible cheaper approximation, but requires the Jacobian in Equation 39 and its own objective checks.

With $\epsilon = 0$, a spike-and-slab prior gives positive probability to an all-zero loading block, where the normalized map is undefined. One must specify its projection as zero at that atom; this yields a discontinuity near zero. Alternatively $\epsilon > 0$ gives a smooth map everywhere with an explicitly scale-dependent regularization. These alternatives are shown for advisor choice, not resolved by silently inserting a floor.

9.4 Nonnegative support and expression–methylation association

Nonnegative score and loading priors imply a nonnegative reconstructed mean $LF_m^\top$. Gaussian noise can be negative, but this does not make that mean family capable of representing arbitrary centered positive and negative signal. Centered expression or methylation M-values and nonnegative priors must therefore be considered together.

Within a single factor, nonnegative loadings in both modalities move in the same direction as its score. They cannot represent opposite directions for that one factor. Multiple complementary factors may produce inverse associations in a fitted dataset, but that is not the same model as a single signed shared effect. With independent nonnegative score priors, the prior marginal cross-feature covariance from shared factors is nonnegative.

Methylation associations depend on genomic context. A primary study reported negative effects enriched near transcription start sites and positive effects enriched in gene bodies (Nature Communications, 2022). The model should not assume a universal inverse association or reverse all CpGs.

Three analysis-scale choices deserve review: retain nonnegative transformed measurements and acknowledge their baseline structure; use signed loading and possibly signed score priors for centered measurements; or extend the model with feature intercepts. Intercepts are **not** part of the core parameterization. If selected, define $y_{mij} = \mu_{mj} + \sum_k l_{ik} f_{mjk} + e_{mij}$ and, at fixed moments, differentiate the Gaussian residual sum of squares to obtain

$$\hat{\mu}_{mj} = \frac{1}{n} \sum_i \left(y_{mij} - \sum_k \bar{l}_{ik} \bar{f}_{mjk} \right).$$

The core updates then use $Y_m - \mu_m$. Score-location constraints and whether the survival projection also uses this centered matrix would need to be specified together. Adding this parameter is an advisor decision, not an undocumented preprocessing substitution.

9.5 Scale, sign, permutation, and factor number

For linear YFB, the simultaneous transformation

$$l'_k = d_k l_k, \quad f'_{mk} = f_{mk}/d_k, \quad \beta'_k = d_k \beta_k, \quad d_k > 0 \quad (41)$$

preserves reconstruction and η . If prior scales are learned, transform them too: $r'_{Lk} = r_{Lk}/d_k$, $r'_{Fmk} = d_k r_{Fmk}$, and $s'_{\beta k} = d_k s_{\beta k}$. The scale ambiguity is not resolved merely by independent scale-adaptive priors.

A reporting convention for a nonzero factor can set $n^{-1} \sum_i l_{ik}^{(2)} = 1$ using $d_k = (n^{-1} \sum_i l_{ik}^{(2)})^{-1/2}$. Transform second moments by d_k^2 for L, β and by d_k^{-2} for F_m , and transform the prior parameters consistently. This is a change of coordinates, not rescaling data. It must not be applied using these same beta rules to the normalized-projection model, where the invariances differ.

Factor permutations always remain. Signed priors permit sign changes with corresponding coefficient/score transformations. General invertible rotations preserve $LF^\top$ but need not preserve independent sparse priors, mean-field distributions, or their objective. Do not enforce orthogonality post hoc and claim the posterior is unchanged.

Report subspace recovery as well as matched-factor recovery. A factor-matching procedure must handle permutation and applicable scale and sign transformations. Keep separate summaries of omics activity, activity in each modality, and prognostic activity; a small beta is not the definition of an absent omics factor. The existing real-data $K_{\text{init}} = 7$ and activity thresholds are not validated multimodal defaults. Numerical K choices remain for the later simulation design.

10 Algorithms and compatibility

10.1 A direct multimodal version of current no-feedback YFB

The closest adaptation preserves the present learning direction:

1. Fit shared L , separate F_m priors, and feature-wise τ_m using the omics objective. The ordinary baseline sets $a_m = 1$.
2. Form $\widehat{Z}$ from the stacked posterior mean loadings using the current global column normalization.
3. Fit β on those fixed projections with the current Cox working curvature and alpha scaling. This can be interleaved with omics iterations for compatibility, but final coefficients must be converged against the final saved projection.

4. Save projection norms, feature order, fitted priors, transformations, and any training-only orientation choice for external prediction.

This is an omics representation plus a conditional survival model. Its objectives are an omics variational objective and a conditional Cox coefficient objective, reported separately. There is no feedback path from survival to the representation. It is not obtained by setting $b = 0$ in the joint target.

When $M = 1$, with the same preprocessing, priors, initialization, update order, floors, and finalization, its algebra reduces to current YFB. The joint uncertainty-aware model need not reproduce that fit: those are different estimation procedures.

10.2 Joint linear-projection variational algorithm

This is the derivation-complete reference algorithm for the raw or fixed-weight linear projection. It uses the full H or the explicitly declared working W ; comparison of these approximations is left to review.

1. **Initialize from training omics.** Use nonzero factor columns and interior prior parameters. Adapt the existing SVD initialization for the selected support; retain initialization provenance. A reconstruction warm start followed by a coefficient fit can reduce early factor collapse. No validation outcomes are consulted.
2. **Set an outer expansion.** Compute all current moments, η_0 , the Cox score, and W . Store t^* and C_0 .
3. **Perform inner coordinate sweeps.** For each factor, update β_k , the shared L_k , and loading coordinates across modalities using Equation 24, Equation 22, and Equation 28. Hold the Cox expansion fixed; update residuals and projected means after each changed coordinate.
4. **Update priors.** Use blockwise EBNM where conditionally separable. For joint loading priors use the fixed-posterior EM steps after fixed-prior scalar sweeps. Compute KL consistently with the current posterior and prior.
5. **Update noise and evaluate.** Apply Equation 30 to each block and evaluate $\mathcal{J}_Q$ using Equation 19 and all KL terms. An exact inner update must not decrease this *fixed-expansion* functional beyond numerical tolerance. A decrease requires diagnosis.
6. **Refresh and monitor.** After inner convergence, refresh the Cox expansion. Monitor coefficient, score, loading, precision, and objective changes, not just training C-index. Record separate reconstruction, Cox, uncertainty, and prior contributions.
7. **Finalize coherently.** Recompute risk projections and all diagnostics from the saved final state. Apply only approved parameterization conventions. Retain nonconvergence, degenerate factors, and boundary priors as explicit diagnostics.

Changing the local expansion can change $\mathcal{J}_Q$ without monotonicity in the exact $\mathcal{J}$. A quadratic Taylor approximation is not a global lower bound, and inner coordinate ascent does not prove outer convergence. For development, approximate $\mathbb{E}\ell_{\text{Cox}}$ with posterior draws and record Monte Carlo uncertainty. Such monitoring can detect failures; it does not turn a local approximation into a theorem. Damping or a proven bound is a later inference choice if outer instability persists. Do not silently fall back to $\alpha_F = 0$ and report joint-model convergence.

10.3 What can be reused

The residual-moment algebra, Normal conjugacy, scalar normal-means posterior machinery, Cox risk-set construction, and most parameter names can be reused. A loading-prior identifier must become a modality-indexed object. L and β remain shared.

The shared score update must aggregate modality contributions. Loading updates require per-modality priors; joint loading updates additionally require the same-factor interaction correction. The joint coefficient and objective calculations require projection uncertainty. Reusing the old combined ELBO monitor unchanged would conceal differences in objective weights and uncertainty.

For the mathematical no-feedback reduction, remove the survival terms from F and treat projections as fixed for coefficient fitting. For the omics-only reduction of the joint model, set $b = 0$; then $q(\beta) = g_\beta$. These are distinct checks.

11 Independent validation and future cohort compatibility

11.1 Frozen training-to-validation map

Let validation data be Y_m^{val} with n_{val} rows. Modality names and within-modality feature identifiers must match the training contract. Reorder subjects by verified identifiers before fitting or prediction, never by row position alone. Validation subjects may be reordered freely provided both modality blocks follow the same permutation.

Feature selection, imputation rules if later approved, transformation parameters, centering/scaling, projection weights, fitted priors, loadings, beta, and factor-retention rules are learned or fixed using training data. Outcome-independent assay QC may operate on validation measurements, but it must not refit the statistical prediction map. All tuning and CV preprocessing must be repeated inside each training fold rather than learned once on the full training cohort.

For the linear projection, the frozen plug-in risk prediction is

$$\hat{\eta}^{\text{val}} = \sum_k \bar{\beta}_k \sum_m c_m Y_m^{\text{val}} \bar{f}_{mk}. \quad (42)$$

For the compatibility model, substitute $\bar{f}_{mk}/D_{kk}^{\text{train}}$ using the saved global training norm. For other approved normalization rules, apply their corresponding frozen training quantities. Normalizing on the validation cohort or refitting beta would change the prediction rule.

Under mean-field joint inference with a linear projection, the plug-in expression is the posterior mean risk predictor. Uncertainty can be propagated by applying Equation 19 with the validation design matrix, without validation outcomes. For stochastic normalized models, posterior-mean prediction instead requires integration through the nonlinear map; normalization of posterior means is an additional approximation.

Missing required modalities or features are not silently imputed by zero and norms are not recomputed on an arbitrary feature subset. For the initial complete-data design, incompatible inputs must stop. An assay-based common feature universe may be prespecified from platform meta-data; outcome- or value-dependent feature selection uses training only. If complete matching is impossible, a missing-feature extension must be reviewed before reporting validation results.

11.2 Latent scores for reconstruction are a separate output

To evaluate validation reconstruction, hold F_m, τ_m, g_{L_k} fixed at training estimates and infer validation score distributions using Equation 22 with validation omics only. Iterate the score coordinates; do not re-estimate the score priors on validation. This produces $\bar{L}^{\text{val}}$ for reconstruction, not the YFB survival covariates.

A point-estimate Gaussian alternative without prior shrinkage solves

$$\hat{L}^{\text{val}} = \left\{ \sum_m a_m Y_m^{\text{val}} \text{diag}(\tau_m) \bar{F}_m \right\} \left\{ \sum_m a_m \bar{F}_m^\top \text{diag}(\tau_m) \bar{F}_m \right\}^+.$$

The pseudoinverse handles rank deficiency. This is weighted least squares conditional on fixed mean loadings, not the same posterior inference as the moment-based score update. It is supplied only to clarify the distinction from the existing LB prediction routine.

Evaluate survival outcomes only after predictions are frozen. Report discrimination and uncertainty alongside convergence and factor activity. Do not use validation concordance to flip the sign of beta. A factor reorientation that preserves η is harmless; negating beta alone changes the statistical fit and is not a joint-objective update. The current training-concordance sign correction belongs to the compatibility baseline and should be reported if used.

11.3 Cohorts are orthogonal to modality partitions

If multiple training cohorts are later included, the observation model can be extended to

$$Y_m = LF_m^\top + CA_m^\top + E_m,$$

where C is an observed $n \times (C_{\text{train}} - 1)$ corner-point cohort-indicator matrix and A_m contains feature-specific offsets. These indicator columns are not biological factors and their survival coefficients are zero in the existing offset extension. Their Gaussian prior updates use the same feature precision and, in a weighted objective, the same modality likelihood weight.

Stratified survival instead sums Equation 11 over training strata and modifies the risk sets. Cohort-specific beta instead replaces β_k by $\beta_k^{c(i)}$ with an explicitly chosen pooling prior. None of these mechanisms partitions F by modality. With one training cohort they are absent; an unseen validation cohort does not supply an estimable new offset or beta without changing the validation design. A full combined-cohort derivation is future work.

12 Translation to implementation and simulation after review

12.1 Minimal interface changes

A future fit should accept a named list of numeric matrices, one per modality, plus survival vectors keyed to verified subject IDs. The fitted object should retain shared EL/EL2 and EBeta/EBeta2 and modality-indexed EF/EF2, Tau, prior fits, feature IDs, and training preprocessing/projection metadata. Existing single-matrix callers should continue to work through their existing path. No such API has been implemented in this deliverable.

Loading-prior arguments must be separate from the shared score prior. The old prior_LF convenience setting can remain an explicit baseline mapping, but it must not cause features from different

modalities to be passed into the same fitted loading prior. Likelihood weights, projection coefficients, and objective type need distinct names and saved values; `alpha_F` must not be overloaded to mean modality balance.

12.2 Computation at CpG scale

At $p_1 = 20,000$ and $p_2 = 400,000$, a dense $p \times p$ double matrix alone would require about 1.41 TB in decimal units. It must not be constructed. A dense $n \times p$ data matrix requires $8np$ bytes (about 1.68 GB at $n = 500$), before residuals, copies, second moments, and temporary products. These are illustrative memory calculations, not selected simulation sample sizes.

Keep modality blocks separate. For diagonal W , scalar loading updates and variance traces need products such as $x_a^\top W x_a$ and $x_a^\top W \bar{z}_{k,-a}$, which can be computed without $X^\top W X$. Updating $\bar{z}_k$ after one loading change costs $O(n)$. Dense diagonal-working-curvature sweeps cost roughly $O(nK \sum_m p_m)$, apart from EBNM optimization. Full-Hessian products can be more expensive; use risk-set products or a declared approximation and benchmark them. Scalar R-to-package calls per CpG may be a practical bottleneck even without a memory problem.

The exact Gauss-Seidel loading updates are sequential through their shared projections. Parallelizing them naively changes the algorithm. Independent simulation seeds and training folds are safer first parallelization targets. If later run on Longleaf, request memory based on measured peak allocation, control BLAS threads inside SLURM workers, and avoid nested parallel oversubscription.

12.3 Synthetic data: mechanism before scenario grid

Generate training and validation subjects independently conditional on common population parameters F_m, τ_m, β and score-prior distributions. Sharing these parameters is what permits evaluation of generalization to new subjects. Independently resampling the validation loading matrix would instead test a changed population and must be labeled a distribution-shift scenario.

For a well-specified YFB experiment, first generate L , then Y_m , then survival using the selected YFB projection of the realized Y_m . The existing multi-cohort simulation uses an $L\beta$ survival mechanism; reusing it without modification would be a misspecification experiment. Include an $L\beta$ mechanism only as an explicitly named robustness comparison.

Candidate scenario dimensions are modality feature-count ratio, sparsity, residual precision, feature redundancy, complementary versus redundant modality signal, nonnegative versus signed true loadings, and factors absent from one modality. Start with small dimensions that preserve the expression-to-CpG ratio, then measure scaling before full-size runs. Exact sample sizes, effect sizes, repetitions, censoring fractions, and factor numbers are intentionally deferred until model review; do not copy the current gene-expression defaults as if validated for multimodal data.

Candidate estimands include shared subspace recovery, matched loading recovery by modality, reconstruction error by modality, survival discrimination on untouched validation data, activity classification, runtime, memory, and convergence/failure rates. Compare expression-only, methylation-only, direct no-feedback multimodal YFB, and the approved joint model using comparable tuning budgets. Assess integration benefit against both single-modality models; report uncertainty over independently generated replicates.

12.4 Real-data preprocessing decisions

The real-data design needs modality-specific assay QC, sample identity checks, feature annotation/version matching, and a frozen training-to-validation transformation. Expression counts may require a library-size-aware transform; methylation proportions may be kept on a bounded measurement scale or transformed to M-values for a Gaussian working likelihood. These choices have different support implications under nonnegative priors. They remain for review with the actual assays and are not implemented by this mathematical note.

Investigate CpG redundancy and batch/platform shifts separately from feature-count balance. Matched subjects do not eliminate assay effects. Do not infer a one-to-one gene-CpG correspondence or a universal inverse relation during feature matching.

13 Verification and advisor decisions

13.1 Reproducible numerical checks

The companion check script validates the algebra on small deterministic examples; it does not fit or simulate a proposed multimodal model. Its examples use nonzero posterior variances and correlated features across modalities so that dropping the interaction or uncertainty terms produces detectable errors.

Verification result: 35 of 35 checks passed.

Algebra or behavior checked	Passed
Breslow score with tied events	TRUE
Full Cox Hessian with tied events	TRUE
Cox Hessian annihilates a constant shift	TRUE
Working curvature is locally at least the Hessian	TRUE
Working curvature is not the exact Hessian diagonal	TRUE
L gradient: full Hessian	TRUE
L curvature: full Hessian	TRUE
Beta gradient with F uncertainty: full Hessian	TRUE
Beta curvature with F uncertainty: full Hessian	TRUE
F gradient with same-factor terms: full Hessian	TRUE
F diagonal curvature: full Hessian	TRUE
Dropping feature interaction is detected: full Hessian	TRUE
L gradient: working diagonal	TRUE
L curvature: working diagonal	TRUE
Beta gradient with F uncertainty: working diagonal	TRUE
Beta curvature with F uncertainty: working diagonal	TRUE
F gradient with same-factor terms: working diagonal	TRUE
F diagonal curvature: working diagonal	TRUE
Dropping feature interaction is detected: working diagonal	TRUE
Risk mean from exact enumeration	TRUE
Risk covariance includes shared-F cross-subject uncertainty	TRUE
Expected full quadratic from exact enumeration	TRUE
Expected squared residual from exact enumeration	TRUE
Weighted Tau score is zero at n / expected SSE	TRUE

Algebra or behavior checked	Passed
Point-normal posterior moments by integration	TRUE
Point-exponential marginal density by integration	TRUE
Point-exponential posterior moments by integration	TRUE
Prior EM mixing, rate, and variance scores	TRUE
EBNM log-normalizer identity equals direct Gaussian KL	TRUE
Unweighted modality sum equals stacked score precision	TRUE
Replicated features preserve inverse-count weighted score sum	TRUE
Feature rescaling preserves precision-weighted mean residual	TRUE
Raw YFB is not invariant to modality rescaling	TRUE
Normalized-projection first derivative	TRUE
Normalized-projection second derivative	TRUE

The checks cover exact Breslow derivatives with tied events; the implemented working-curvature relation; finite-difference derivatives of the expected quadratic for L, F, β, τ ; enumerated risk moments; prior posterior moments and hyperparameter stationarity; modality-partition reduction; weighted duplication sums; and normalized-projection derivatives.

These checks verify local formulas, not convergence or predictive performance of a future implementation. No numerical fit comparison is claimed. After code implementation, run the repository test suite and add end-to-end tests for subject/feature alignment, frozen prediction, separate prior fitting, objective accounting, and the matched single-modality reduction.

13.2 Decisions to take to the advisors

Decision	Starting point	Alternative or required resolution
Representation	One shared L ; separate F_m priors	Allow a block to shrink to zero; already agreed
Data structure	One matched training cohort and one matched validation cohort	Multiple training cohorts deferred
Observation model	Gaussian working likelihood; feature-wise precision MLE	Confirm transforms and residual adequacy
Loading support	Nonnegative point-exponential	Compare signed loadings and centered-data compatibility
Evidence balance	Ordinary observation likelihood first	Explicit modality weights if justified
Projection	Current normalized mean projection for compatibility	Linear joint reference versus nonlinear normalized joint model
Supervision	Joint survival-informed F derivation	Retain no-feedback YFB as the direct implementation baseline

Decision	Starting point	Alternative or required resolution
Inference	Coordinate updates for a declared Cox surrogate	Select working versus exact curvature and stability strategy
Beta prior	Existing zero-centered Normal per factor	Pooling or sparse alternative requires separate choice
Validation	Frozen training map; outcomes only for evaluation	Missing-feature or cohort adaptation needs a new protocol

The main dependency is projection choice: it determines whether normal-means coordinate updates remain available. Weighting and prior support must then be assessed on that same parameterization and analysis scale. This document supplies the mathematical alternatives; it does not approve one by implication.

13.3 Build, provenance, and teach-back

From the repository root:

```
Rscript derivations/multimodal_YFB/verify_derivation.R
quarto render derivations/multimodal_YFB/multimodal_YFB_derivation.qmd --to pdf
```

Rendering executes the verification script and stops on a failed assertion. The source remains editable alongside the PDF. The standalone check script uses base R; rendering additionally needs Quarto, knitr, and a working LaTeX installation. It does not source the production fitting scripts or load real data.

The mathematical document’s input is the existing YFB parameterization and the agreed matched-modality design; its output is a set of explicit estimation objectives and derived update procedures. The main design choice is to separate direct compatibility from joint inference so that new modality priors do not conceal changes in survival supervision. The main limitation is that a local Cox surrogate and a mean-field approximation do not establish global convergence or calibrated uncertainty.

The check script’s input is small deterministic numerical fixtures. It compares derivatives, exact finite-support expectations, and numerical integrals with the equations. Its output is a named table of assertions; any failure terminates execution. These computations test mathematical identities rather than performance of an unimplemented model.

14 Sources and local references

The update algebra in this note is derived explicitly above. The following sources establish the empirical-Bayes framework and software conventions; they are not substitutes for the derivation.

- Wang and Stephens (2021), [Empirical Bayes Matrix Factorization](#). Background on learning factor/loading priors with variational empirical Bayes.

- Willwerscheid, Carbonetto, and Stephens (2025), [ebnm: An R Package for Solving the Empirical Bayes Normal Means Problem Using a Variety of Prior Families](#). Background on normal-means prior estimation.
- [Official ebnm reference](#). Fixed prior mode, prior scale conventions, fixed-prior posterior calls, and posterior moment outputs. Local package version inspected: 1.1.42.
- [Quantifying the role of transcript levels in mediating DNA methylation effects on complex traits and diseases](#) (Nature Communications, 2022). Biological context for the sign comparison.
- Local mathematical references: `derivations/cox_on_YF/` and `docs/notes/YFB_update_derivations_detailed_05_08_26.qmd`. These describe earlier YFB formulas; the audit in Section 2 identifies where they cannot be carried over unchanged.
- Local implementation references: `code/fit_cox_on_yf.R`; `code/update_L_surv_YFB.R`; `code/update_F_surv_YFB.R`; `code/update_beta.R`; `code/update_tau.R`; `code/compute_elbo.R`; `code/predict_cox_on_yf.R`.
- Cohort references: `code/update_F_cohort.R` and `code/update_beta_cohort.R`; stratified risk-set handling in `code/fit_cox_on_yf.R`.